

SBN Explorer

Design Space and Independence of Architectures

1 Design Space

S – Stratification. The network is said to be stratified in the SPN sense when there exist an integer $d \geq 2$, a layer map $\ell : V \rightarrow \{1, \dots, d\}$, and a type map $\tau : \{1, \dots, d\} \rightarrow \{\mathbf{S}, \mathbf{P}\}$, such that $\tau(k+1) \neq \tau(k) \quad \forall k \in \{1, \dots, d-1\}$, and, for every vertex $v \in V$,

$$v \in \ell^{-1}(k) \implies f_v \text{ is of type } \tau(k).$$

Thus, stratification refers here to an alternating substitution–permutation layer structure, rather than to a feed-forward constraint on edges. In particular, this notion of stratification does not imply acyclicity. No constraint is imposed on the edge set with respect to the layer order.

A – Acyclicity. The network is acyclic when its structural graph has no directed cycle:

$$A(\mathcal{C}) = 1 \iff G = (V, E) \text{ is a directed acyclic graph.}$$

If $A(\mathcal{C}) = 0$, the graph may contain feedback loops, and the SBN is then understood dynamically through parallel iteration of its global update map.

R – Regularity. The network is regular when all input-to-output minimal propagation times coincide:

$$R(\mathcal{C}) = 1 \iff \exists T \in \mathbb{N} \text{ such that } \forall x \in V_{\text{in}}, \forall y \in V_{\text{out}},$$

$$\tau_G(x, y) := \min\{t \geq 0 \mid y \text{ depends on } x \text{ after } t \text{ iterations}\}.$$

In the feed-forward case, when V_{in} and V_{out} are the source and sink sets, this reduces to the classical equal-depth condition. In the recurrent case, it remains meaningful as a purely dynamical equal-delay condition.

I – Interleaving. The network is interleaved when at least one edge crosses the prescribed block decomposition:

$$I(\mathcal{C}) = 1 \iff \exists (u, v) \in E \text{ such that } b(u) \neq b(v).$$

where $b : V \rightarrow \{1, \dots, B\}$ is a fixed block map. Equivalently,

$$I(\mathcal{C}) = 0 \iff \forall (u, v) \in E, \quad b(u) = b(v).$$

Thus I measures cross-block coupling independently of layering, acyclicity, or locality.

H – Homogeneity. The network is homogeneous when each prescribed layer carries a unique local rule up to permutation of inputs:

$$H(\mathcal{C}) = 1 \iff \forall k \in \{1, \dots, d\}, \exists r_k \in \mathbb{N}, \exists \varphi_k : \{0, 1\}^{r_k} \rightarrow \{0, 1\}$$

such that every vertex $v \in V_k$ has indegree r_k and

$$\forall v \in V_k, \quad f_v = \varphi_k \circ P_v$$

for some permutation P_v of the r_k input coordinates. Hence H is a symmetry condition on local update rules relative to the fixed layer partition, not a consequence of S .

L – Locality. The network is local when every interaction stays within the prescribed geometric radius:

$$L(\mathcal{C}) = 1 \iff \forall (u, v) \in E, \quad \rho(u, v) \leq \delta_0.$$

where (V, ρ) is a fixed metric space. Because δ_0 is fixed in advance, L is a genuine design predicate and not a vacuous bounded-diameter statement.

2 Logical Independence of Architectures

The family $\Omega := \{S, A, R, I, H, L\}$ is *logically independent*.

Proposition 1 (Logical independence). *Let $\Omega = \{S, A, R, I, H, L\}$ be the six predicates defined on enriched SBN. Then they are logically orthogonal in the following sense: for every $X \in \Omega$, there exist two enriched SBNs $\mathcal{C}_X^+$ and $\mathcal{C}_X^-$ such that $X(\mathcal{C}_X^+) = 1$, $X(\mathcal{C}_X^-) = 0$, while $Y(\mathcal{C}_X^+) = Y(\mathcal{C}_X^-)$ for all $Y \neq X$. Consequently, no predicate among Ω is logically implied by the conjunction of the other five.*

Proof. We write the predicate vector of an enriched SBN as $\nu(\mathcal{C}) = (S, A, R, I, H, L) \in \mathbb{B}^6$. For each coordinate, we exhibit two witnesses differing only on that coordinate.

1. Toggling S . Take $V = \{a, b, c\}$, $E = \{a \rightarrow b, b \rightarrow c\}$, with layer map $\ell(a) = 1$, $\ell(b) = 2$, $\ell(c) = 3$, boundary sets $V_{\text{in}} = \{a\}$, $V_{\text{out}} = \{c\}$, one single block, and a metric such that all used edges are local.

For $\mathcal{C}_S^+$, choose an alternating type map $\tau(1) = S$, $\tau(2) = P$, $\tau(3) = S$, and choose local rules consistent with these layer types.

For $\mathcal{C}_S^-$, keep the same graph, same layer map, same blocks, same metric, but choose a non-alternating typing, for instance $\tau(1) = S$, $\tau(2) = S$, $\tau(3) = P$, again with local rules consistent with these assigned layer types.

Then $S(\mathcal{C}_S^+) = 1$, $S(\mathcal{C}_S^-) = 0$, while all other predicates are unchanged: $A = 1$, $R = 1$, $I = 0$, $H = 1$, $L = 1$. Hence

$$\nu(\mathcal{C}_S^+) = (1, 1, 1, 0, 1, 1), \quad \nu(\mathcal{C}_S^-) = (0, 1, 1, 0, 1, 1).$$

2. Toggling A. Take $V = \{a, b\}$, with $\ell(a) = 1, \ell(b) = 2, \tau(1) = S, \tau(2) = P$, $V_{\text{in}} = \{a\}, V_{\text{out}} = \{b\}$, one block, and local metric.

Let $E^+ = \{a \rightarrow b\}$, $E^- = \{a \rightarrow b, b \rightarrow a\}$. Then $T = 1$ in both cases, since the alternating layer typing is unchanged and does not constrain edge orientation. However, $A(\mathcal{C}_A^+) = 1, A(\mathcal{C}_A^-) = 0$. All other predicates remain unchanged: $R = 1, I = 0, H = 1, L = 1$. Thus

$$\nu(\mathcal{C}_A^+) = (1, 1, 1, 0, 1, 1), \quad \nu(\mathcal{C}_A^-) = (1, 0, 1, 0, 1, 1).$$

3. Toggling R. For $R = 1$, take vertices $a \in V_1, b, d \in V_2, c, e \in V_3$, with $\tau(1) = S, \tau(2) = P, \tau(3) = S, V_{\text{in}} = \{a\}, V_{\text{out}} = \{c, e\}$, one block, homogeneous local rules inside each layer, and local metric. Let $E^+ = \{a \rightarrow b, b \rightarrow c, a \rightarrow d, d \rightarrow e\}$. Then $\tau_G(a, c) = \tau_G(a, e) = 2$, so $R = 1$. Also $S = 1, A = 1, I = 0, H = 1, L = 1$.

For $R = 0$, take vertices $a \in V_1, b, d \in V_2, e \in V_3$, with the same alternating typing $\tau(1) = S, \tau(2) = P, \tau(3) = S, V_{\text{in}} = \{a\}, V_{\text{out}} = \{d, e\}$, and $E^- = \{a \rightarrow d, a \rightarrow b, b \rightarrow e\}$. Then $\tau_G(a, d) = 1, \tau_G(a, e) = 2$, hence $R = 0$, while still $S = 1, A = 1, I = 0, H = 1, L = 1$. Therefore

$$\nu(\mathcal{C}_R^+) = (1, 1, 1, 0, 1, 1), \quad \nu(\mathcal{C}_R^-) = (1, 1, 0, 0, 1, 1).$$

4. Toggling I. Take the same graph $a \rightarrow b \rightarrow c$ with $\ell(a) = 1, \ell(b) = 2, \ell(c) = 3, \tau(1) = S, \tau(2) = P, \tau(3) = S, V_{\text{in}} = \{a\}, V_{\text{out}} = \{c\}$, and local metric.

For $I = 0$, choose one block: $b(a) = b(b) = b(c) = 1$. For $I = 1$, choose for instance $b(a) = 1, b(b) = 2, b(c) = 2$. Then the edge $a \rightarrow b$ is cross-block, so $I = 1$. In both cases, $S = 1, A = 1, R = 1, H = 1, L = 1$. Thus

$$\nu(\mathcal{C}_I^-) = (1, 1, 1, 0, 1, 1), \quad \nu(\mathcal{C}_I^+) = (1, 1, 1, 1, 1, 1).$$

5. Toggling H. Take $a \in V_1, b, c \in V_2, d \in V_3$, with graph $E = \{a \rightarrow b, a \rightarrow c, b \rightarrow d, c \rightarrow d\}$, one block, $\tau(1) = S, \tau(2) = P, \tau(3) = S, V_{\text{in}} = \{a\}, V_{\text{out}} = \{d\}$, and local metric. Then $S = 1, A = 1, R = 1, I = 0, L = 1$.

For $H = 1$, choose identical affine unary rules on b and c , for instance $f_b(x) = x, f_c(x) = x$. For $H = 0$, choose non-equivalent affine unary rules, e.g. $f_b(x) = x, f_c(x) = 1 + x$. These two rules are both of type P, so S remains equal to 1, but they are not equivalent up to input permutation. Hence

$$\nu(\mathcal{C}_H^+) = (1, 1, 1, 0, 1, 1), \quad \nu(\mathcal{C}_H^-) = (1, 1, 1, 0, 0, 1).$$

6. Toggling L. Take again the graph $a \rightarrow b \rightarrow c$, with one block and singleton layers, so that $\ell(a) = 1, \ell(b) = 2, \ell(c) = 3, \tau(1) = S, \tau(2) = P, \tau(3) = S$, and therefore $S = 1, A = 1, R = 1, I = 0, H = 1$. Keep the graph, block map, layer map, type map, and local rules fixed. For $L = 1$, choose a metric ρ^+ and radius δ_0 such that $\rho^+(a, b) \leq \delta_0, \rho^+(b, c) \leq \delta_0$. For $L = 0$, choose another metric ρ^- with $\rho^-(a, b) > \delta_0, \rho^-(b, c) \leq \delta_0$. Then $L(\mathcal{C}_L^+) = 1, L(\mathcal{C}_L^-) = 0$, while the other five predicates are unchanged:

$$\nu(\mathcal{C}_L^+) = (1, 1, 1, 0, 1, 1), \quad \nu(\mathcal{C}_L^-) = (1, 1, 1, 0, 1, 0).$$

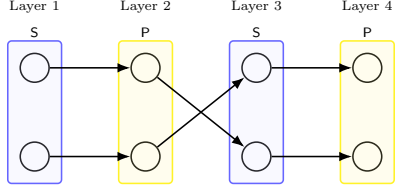
Collecting the six witness pairs proves the claim. $\square$

Corollary 1. *Since the six predicates are logically independent, their joint valuation spans the Boolean cube $\{0,1\}^6$. Hence the architectural design space contains $2^6 = 64$ distinct classes.*

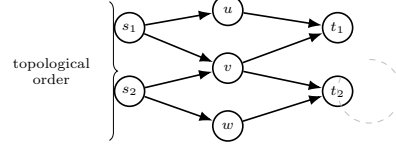
Corollary 2. *The five predicates $\Omega^- := \{A, R, I, H, L\}$ are also logically independent.*

3 Intuition and visualization

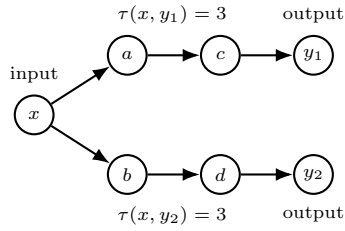
Figure 1 depicts the intuitive idea behind the structural constraints.



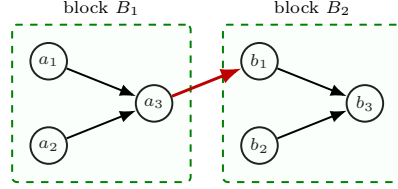
[S] **Stratification.** Layers alternate between nonlinear substitution type (S) and linear/affine permutation–diffusion type (P).



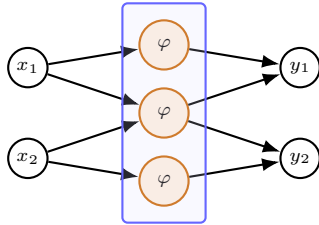
[A] **Acyclicity.** The graph admits a topological order: arrows always move forward in the causal ordering, and no structural feedback loop is present.



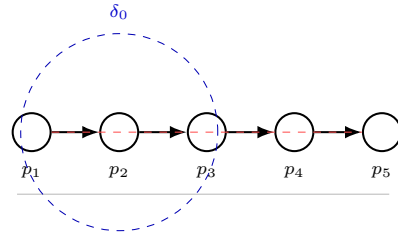
[R] **Regularity.** All input-output minimal propagation times are equal. Regularity removes latency bias between distinct routes through the network.



[I] **Interleaving.** At least one dependency crosses the prescribed block decomposition. The highlighted edge couples two otherwise distinct modules.



[H] **Homogeneity.** Inside a given layer, all vertices implement the same Boolean rule (up to permutation of input wires).



[L] **Locality.** Every edge must stay within the prescribed geometric radius δ_0 . The grey dashed arrow indicates a non-local shortcut excluded by the constraint.

Figure 1: Intuitive graph sketches for the six orthogonal predicates — Each panel shows a canonical graph pattern illustrating the meaning of the predicate when it is satisfied.